

8.3 Characterizations of estimability

Theorem 8.2.2 reduced estimability to a single condition, $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. In practice that condition appears in several forms, depending on what is at hand. We may have the rows of $\mathbf{X}$, the matrix $\mathbf{X}^\top \mathbf{X}$ and one of its generalized inverses, a collection of least squares solutions, or an SVD. This section collects the equivalent forms in one theorem, derives the estimator of an estimable function with its variance, and then turns to the practical question of deciding estimability in floating-point arithmetic, where exact rank is not available.

8.3.1 Eight equivalent conditions

Theorem 8.3.1. Characterizations of estimability

Let $\mathbf{X}$ be $n \times p$, let $\mathbf{G}$ denote a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, and let $\lambda \in \mathbb{R}^p$. The following are equivalent:

- (a) $\lambda^\top \beta$ is estimable;
- (b) $\lambda \in \mathcal{C}(\mathbf{X}^\top)$: there is $\rho \in \mathbb{R}^n$ with $\lambda^\top = \rho^\top \mathbf{X}$;
- (c) $\lambda \perp \mathcal{N}(\mathbf{X})$: $\lambda^\top \mathbf{v} = 0$ whenever $\mathbf{X}\mathbf{v} = \mathbf{0}$;
- (d) $\lambda \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$: the system $\mathbf{X}^\top \mathbf{X}\mathbf{r} = \lambda$ has a solution;
- (e) $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ for some generalized inverse $\mathbf{G}$;
- (f) $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ for every generalized inverse $\mathbf{G}$;
- (g) for every $\mathbf{y}$, $\lambda^\top \mathbf{b}$ has the same value for all solutions $\mathbf{b}$ of the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ (Theorem 6.4.2);
- (h) $\text{rank} \begin{pmatrix} \mathbf{X} \\ \lambda^\top \end{pmatrix} = \text{rank}(\mathbf{X})$.

Proof. (a) $\Leftrightarrow$ (b) is Theorem 8.2.2. (b) $\Leftrightarrow$ (c) holds because $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ (Corollary 6.2.4(d)). (b) $\Leftrightarrow$ (d) holds because $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ (Corollary 6.2.4(b)). (d) $\Rightarrow$ (f): if $\lambda = \mathbf{X}^\top \mathbf{X}\mathbf{r}$, then for every generalized inverse $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{r}^\top (\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X}) = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} = \lambda^\top$. (f) $\Rightarrow$ (e) holds because generalized inverses exist (Theorem 1.3.2). (e) $\Rightarrow$ (d): transposing $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$ gives $\mathbf{X}^\top \mathbf{X}(\mathbf{G}^\top \lambda) = \lambda$, so $\mathbf{r} = \mathbf{G}^\top \lambda$ solves the system. (b) $\Leftrightarrow$ (g) is Theorem 6.4.4. (b) $\Leftrightarrow$ (h): the rank of the stacked matrix is the dimension of its row space, $\mathcal{C}([\mathbf{X}^\top, \lambda])$. This contains $\mathcal{C}(\mathbf{X}^\top)$, and it has the same dimension iff it equals $\mathcal{C}(\mathbf{X}^\top)$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. ■

Conditions (e) and (f) are Corollary 1.4.2 applied to the normal equations. Condition (g) is the one that matters for interpretation: an estimable function is one whose estimate does not depend on the arbitrary choice of solution. Condition (c) is the one to use when proving that something is *not* estimable, since a single null vector $\mathbf{v}$ with $\lambda^\top \mathbf{v} \neq 0$ settles the question. Condition (h) is the most direct computational test, but in floating point it inherits all the difficulties of deciding a rank.

8.3.2 The estimator and its variance

For an estimable function, every least squares solution produces the same estimate, and that estimate has a clean description.

Proposition 8.3.2. Least squares estimation of estimable functions

Let $\lambda^\top \beta$ be estimable, with $\lambda = \mathbf{X}^\top \rho = \mathbf{X}^\top \mathbf{X} \mathbf{r}$, let $\hat{\beta}$ be any least squares solution and $\mathbf{G}$ any generalized inverse of $\mathbf{X}^\top \mathbf{X}$. Then:

- (a) $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M} \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$;
- (b) $E(\lambda^\top \hat{\beta}) = \lambda^\top \beta$;
- (c) if $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda = \sigma^2 \mathbf{r}^\top \lambda = \sigma^2 \|\mathbf{M} \rho\|^2,$$

the same for every $\mathbf{G}$;

- (d) more generally, if $\Lambda^\top \beta$ is estimable, then $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$ for every $\mathbf{G}$.

Proof.

- (a) The first equality is Theorem 6.4.4. For the second, the normal equations give $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} = \lambda^\top \hat{\beta}$. The last is the second with the solution $\hat{\beta} = \mathbf{G} \mathbf{X}^\top \mathbf{y}$ and $\mathbf{r} = \mathbf{G}^\top \lambda$ (see the proof of Theorem 8.3.1). (b)–(d) were proved in Chapter 7: unbiasedness and $\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda$ are Theorem 7.2.1(a), and the matrix form $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$ is part (d) of the same theorem. Neither depends on $\mathbf{G}$, because $\mathbf{M} = \mathbf{X} \mathbf{G} \mathbf{X}^\top$ for every $\mathbf{G}$ (Theorem 6.3.6). The second form in (c) follows from $\lambda^\top \mathbf{G} \lambda = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \lambda$. The third is $\rho^\top \mathbf{M} \rho = \|\mathbf{M} \rho\|^2$. ■

The formula $\sigma^2 \lambda^\top \mathbf{G} \lambda$ is the rank-deficient version of the familiar $\sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ (Theorem 5.5.1). The generalized inverse is arbitrary, but the quadratic form in an estimable λ is not. The condition $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ under which Chapter 7 proved optimality is exactly estimability, so the Gauss–Markov theorem and the normal theory (Theorem 7.5.1) apply to every estimable $\lambda^\top \hat{\beta}$ unchanged.

8.3.3 What a generalized inverse reports for a non-estimable function

Software that solves the normal equations with a particular generalized inverse will print $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$ and $\hat{\sigma}^2 \lambda^\top \mathbf{G} \lambda$ for any λ whatever. To see what these numbers mean when λ is not estimable, look at the matrix

$$\mathbf{H}_\mathbf{G} = \mathbf{G} \mathbf{X}^\top \mathbf{X}. \quad (8.3.1)$$

Proposition 8.3.3. The matrix $\mathbf{GX}^\top\mathbf{X}$

For any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top\mathbf{X}$, with $\mathbf{H}_\mathbf{G}$ as in Eq. 8.3.1:

- (a) $\mathbf{H}_\mathbf{G}$ is idempotent, $\mathcal{N}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{X})$, $\text{rank}(\mathbf{H}_\mathbf{G}) = r$, and the rows of $\mathbf{H}_\mathbf{G}$ span $\mathcal{C}(\mathbf{X}^\top)$;
- (b) $\lambda^\top\boldsymbol{\beta}$ is estimable iff $\lambda^\top\mathbf{H}_\mathbf{G} = \lambda^\top$;
- (c) the solution $\hat{\boldsymbol{\beta}}_\mathbf{G} = \mathbf{GX}^\top\mathbf{Y}$ has $E(\hat{\boldsymbol{\beta}}_\mathbf{G}) = \mathbf{H}_\mathbf{G}\boldsymbol{\beta}$. So for *every* λ , $\lambda^\top\hat{\boldsymbol{\beta}}_\mathbf{G}$ is unbiased for the estimable function $\lambda^\top\mathbf{H}_\mathbf{G}\boldsymbol{\beta}$, which equals $\lambda^\top\boldsymbol{\beta}$ iff $\lambda^\top\boldsymbol{\beta}$ is estimable;
- (d) for the Moore–Penrose inverse $\mathbf{G} = (\mathbf{X}^\top\mathbf{X})^+$, $\mathbf{H}_\mathbf{G}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$.

Proof.

- (a) $\mathbf{H}_\mathbf{G}^2 = \mathbf{G}(\mathbf{X}^\top\mathbf{X}\mathbf{G}\mathbf{X}^\top\mathbf{X}) = \mathbf{GX}^\top\mathbf{X} = \mathbf{H}_\mathbf{G}$. By Proposition 1.3.3(a), $\mathcal{N}(\mathbf{GX}^\top\mathbf{X}) = \mathcal{N}(\mathbf{X}^\top\mathbf{X})$, which is $\mathcal{N}(\mathbf{X})$ by Corollary 6.2.4(a). Rank–nullity gives rank r . The row space of $\mathbf{H}_\mathbf{G}$ is $\mathcal{C}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{H}_\mathbf{G})^\perp = \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$.
- (b) is Theorem 8.3.1(e),(f).
- (c) $E(\mathbf{GX}^\top\mathbf{Y}) = \mathbf{GX}^\top\mathbf{X}\boldsymbol{\beta}$. The coefficient vector $\mathbf{H}_\mathbf{G}^\top\lambda$ lies in the row space of $\mathbf{H}_\mathbf{G}$, which is $\mathcal{C}(\mathbf{X}^\top)$, so the function is estimable. It equals $\lambda^\top\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ iff $\mathbf{H}_\mathbf{G}^\top\lambda = \lambda$, which is (b).
- (d) By the Penrose conditions (Definition 1.3.2), $(\mathbf{X}^\top\mathbf{X})^+\mathbf{X}^\top\mathbf{X}$ is symmetric and idempotent. It is therefore an orthogonal projection (Theorem 6.3.3), and its null space is $\mathcal{N}(\mathbf{X})$ by (a). So it projects onto $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$. ■

Part (c) answers the question. For a nonestimable λ , the printed number is a perfectly good unbiased estimate, but of a *different* function, $\lambda^\top\mathbf{H}_\mathbf{G}\boldsymbol{\beta}$, and which function depends on the software’s choice of $\mathbf{G}$. The printed “variance” $\sigma^2\lambda^\top\mathbf{G}\lambda$ is the variance of that statistic when $\mathbf{G}$ is symmetric and reflexive, as the Moore–Penrose inverse is, but in general it need not be the variance of anything (Exercise C1). Part (d) gives the cleanest picture. With the Moore–Penrose inverse, $\boldsymbol{\beta} = \mathbf{H}\boldsymbol{\beta} + (\mathbf{I} - \mathbf{H})\boldsymbol{\beta}$ splits the coefficient vector into its component in the estimable space, which the data determine, and its component in $\mathcal{N}(\mathbf{X})$, which they cannot see. The minimum-norm solution estimates the first component and sets the second to zero.

Example. Checking estimability in a one-way layout

Eighteen observations fall into four groups of sizes 4, 6, 3, 5, with $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4]$ and $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$, so $\text{rank}(\mathbf{X}) = 4$ and $\mathcal{N}(\mathbf{X})$ is spanned by $(1, -1, -1, -1, -1)^\top$. By condition (c), $\lambda^\top\boldsymbol{\beta}$ is estimable iff $\lambda_0 = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$. The listing tests five vectors in two ways: condition (c) through the SVD of $\mathbf{X}$, and condition (e) with three generalized inverses (Moore–Penrose, the one that deletes the intercept row and column, and a non-symmetric one built as in Proposition 1.3.3(e)). The two tests always agree, and condition (e) holds for all three inverses or for none, as (f) requires.

$\lambda^\top\boldsymbol{\beta}$		$\lambda^\top\mathbf{G}\lambda$:		
		Moore–Penrose	drop intercept	non-symmetric
$\mu + \alpha_1$	yes	0.250	0.250	0.250

$\alpha_1 - \alpha_2$	yes	0.417	0.417	0.417
$\alpha_1 + \alpha_2 - 2\alpha_3$	yes	1.750	1.750	1.750
α_1	no	0.188	0.250	0.630
$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$	no	0.038	0.950	2.767

For the estimable rows the quadratic form is the same for all three and matches the direct computation: $\text{Var}(\bar{y}_1 - \bar{y}_2) = \sigma^2(\frac{1}{4} + \frac{1}{6})$, for instance. For α_1 the “variances” disagree. With the drop-intercept inverse, the solution is $\hat{\beta}_{\mathbf{G}} = (0, \bar{y}_1, \dots, \bar{y}_4)^\top$, so the reported “ $\hat{\alpha}_1$ ” is $\bar{y}_1$, which estimates $\mu + \alpha_1$. Its reported variance $0.250\sigma^2$ is the variance of $\bar{y}_1$, which is correct, but $\bar{y}_1$ estimates $\mu + \alpha_1$, not α_1 .

```
import numpy as np

groups = np.repeat([0, 1, 2, 3], [4, 6, 3, 5])          # four groups, sizes 4, 6, 3,
5
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # [1, z1, z2, z3, z4
]
A = X.T @ X
```

[Run this code online](#)

```
def is_estimable(lam, X, tol=None):
    """lambda is estimable iff it is orthogonal to N(X); use the SVD of X."""
    U, s, Vt = np.linalg.svd(X)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0]
    r = int(np.sum(s > tol))
    V0 = Vt[r:].T                                     # orthonormal basis of N(X)
    return np.linalg.norm(V0.T @ lam) <= 1e-8 * max(1.0, np.linalg.norm(lam))

G_mp = np.linalg.pinv(A)                             # Moore-Penrose inverse of X^T X
G_ref = np.zeros((5, 5))
G_ref[1:, 1:] = np.linalg.inv(A[1:, 1:])             # delete the intercept row and
column
rng = np.random.default_rng(3)
G_odd = G_mp + (np.eye(5) - G_mp @ A) @ rng.normal(size=(5, 5)) # non-symmetric g-
inverse

candidates = {"mu + alpha_1": [1, 1, 0, 0, 0],
              "alpha_1 - alpha_2": [0, 1, -1, 0, 0],
              "alpha_1 + alpha_2 - 2 alpha_3": [0, 1, 1, -2, 0],
              "alpha_1": [0, 1, 0, 0, 0],
              "alpha_1 + ... + alpha_4": [0, 1, 1, 1, 1]}
for name, lam in candidates.items():
    lam = np.array(lam, dtype=float)
    H_checks = [np.allclose(lam @ G @ A, lam) for G in (G_mp, G_ref, G_odd)]
    variances = [lam @ G @ lam for G in (G_mp, G_ref, G_odd)]
    print(f"{name:30s} SVD test: {is_estimable(lam, X)!s:5s}",
          f" lambda'GX'X = lambda': {H_checks}",
          f" lambda'G lambda: {np.round(variances, 3)}")
```

[Run this code online](#)

8.3.4 Deciding estimability in floating point

Every condition in [Theorem 8.3.1](#) is a statement of exact linear algebra. On a computer, the rank of $\mathbf{X}$ must be decided with a tolerance, and the answer to “is λ estimable?” inherits that tolerance. The most reliable test works with $\mathbf{X}$ itself, not with $\mathbf{X}^\top \mathbf{X}$. Compute the singular value decomposition $\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^\top$. Declare the singular values below a threshold τ to be zero. With the usual choice $\tau = \max(n, p) \epsilon d_1$, where d_1 is the largest singular value and $\epsilon \approx 2.2 \times 10^{-16}$ is the machine precision, the columns $\mathbf{V}_0$ of $\mathbf{V}$ that belong to the discarded singular values form an orthonormal basis of the numerical null space. Then λ is declared estimable iff $\|\mathbf{V}_0^\top \lambda\|$ is negligible relative to $\|\lambda\|$. This is condition (c), and it is what the function `is_estimable` in the listing does. [Chapter 10](#) treats the SVD and its rank decisions in detail (Theorem 10.4.1).

Two things can go wrong, and both matter.

Exact dependence can be blurred by rounding. If a column is computed, say as a sum of other columns, rounding may leave a singular value of order 10^{-15} rather than zero. The tolerance exists to absorb this.

Near dependence is not dependence. If the columns are nearly but not exactly dependent, every λ is estimable in the exact sense, but some are estimable only with enormous variance. Estimability is then a matter of degree, not a yes-or-no property.

Example. Estimable in principle, not in practice

Take twelve equally spaced values t_i on $[-1, 1]$ and the model matrix with columns $\mathbf{1}$, $\mathbf{t}$ and $\mathbf{t} + \delta \mathbf{w}$, where $w_i = \cos(\pi t_i)$ and δ is a small perturbation. At $\delta = 0$ the last two columns coincide. Then $\beta_1 + \beta_2$ is estimable and β_1 is not. For $\delta > 0$ the matrix has full rank, so everything is estimable. But [Figure 8.3.1](#) shows how the variance of $\hat{\beta}_1$ grows as $\delta \rightarrow 0$: in proportion to $1/\delta^2$ (the fitted slope on the log–log scale is -2.00). At $\delta = 10^{-4}$, the smallest singular value of $\mathbf{X}$ is 1.8×10^{-4} , and $\text{Var}(\hat{\beta}_1) = 1.56 \times 10^7 \sigma^2$, while $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$ is unaffected. The function that is estimable at $\delta = 0$ stays well estimated, and the one that is not becomes useless long before it becomes nonestimable.

The listing adds a warning about computation. At $\delta = 10^{-8}$, the SVD of $\mathbf{X}$ still sees three nonzero singular values, so β_1 is (barely) estimable. But the matrix $\mathbf{X}^\top \mathbf{X}$ has eigenvalues equal to the *squared* singular values. Its smallest eigenvalue is about 3×10^{-16} , below any sensible tolerance, and `pinv(X.T @ X)` silently treats $\mathbf{X}^\top \mathbf{X}$ as having rank 2. It then reports 0.0529 as the “variance” of $\hat{\beta}_1$, which is the Moore–Penrose quadratic form of a nonestimable function. The true variance, computed from the SVD of $\mathbf{X}$ itself, is more than 10^{16} times larger. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number, as [Section 6.10](#) showed, and here that turns a nearly nonestimable function into a numerically nonestimable one.

```
t = np.linspace(-1, 1, 12)
w = np.cos(np.pi * t)                                     # a fixed direction to perturb
                        along
def near_design(delta):
    """Columns 1, t and t + delta*w: exactly rank deficient at delta = 0."""
    return np.column_stack([np.ones_like(t), t, t + delta * w])

for delta in [1.0, 1e-2, 1e-4, 1e-8, 0.0]:
    Xd = near_design(delta)
    s = np.linalg.svd(Xd, compute_uv=False)
    G = np.linalg.pinv(Xd.T @ Xd)
    v_sum = np.array([0, 1, 1]) @ G @ [0, 1, 1]           # beta_1 + beta_2
```

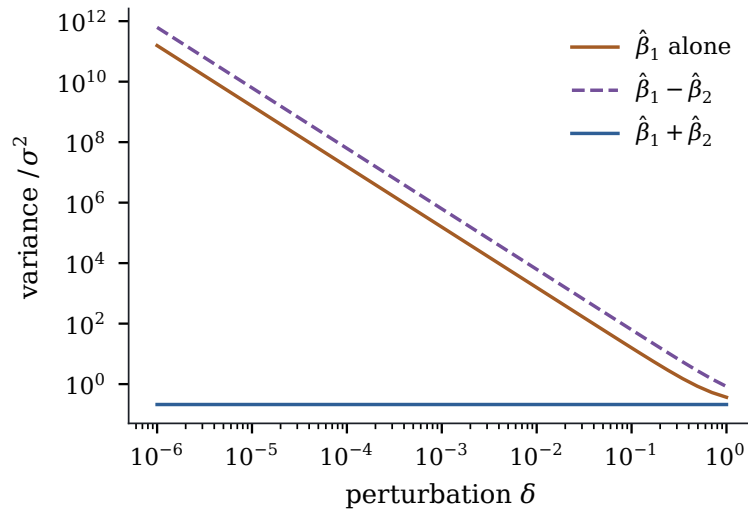


Figure 8.3.1: Variances of three linear functions of $\hat{\beta}$ in the design with columns $\mathbf{1}, \mathbf{t}, \mathbf{t} + \delta \mathbf{w}$. As $\delta \rightarrow 0$, the functions that are not estimable at $\delta = 0$ (β_1 alone and $\beta_1 - \beta_2$) have variances growing like δ^{-2} , while $\beta_1 + \beta_2$, which is estimable at $\delta = 0$, is unaffected.

```
v_one = np.array([0, 1, 0]) @ G @ [0, 1, 0]      # beta_1 alone
print(f"delta={delta:7.0e}  smallest singular value {s[-1]:8.1e}",
      f" Var(b1+b2)/s2 = {v_sum:8.3f}  Var(b1)/s2 = {v_one:9.2e}",
      f" beta_1 estimable? {is_estimable(np.array([0., 1, 0]), Xd)}")
```

[Run this code online](#)

The practical rule has two parts. Decide estimability with the SVD of $\mathbf{X}$, never with $\mathbf{X}^\top \mathbf{X}$. Then look at the variance, not only the verdict. A function that passes the test with a variance of $10^7 \sigma^2$ has, for practical purposes, not been estimated. Collinearity, in [Chapter 26](#), is the study of this middle ground.

8.3.5 Exercises

8.3.5.1 A. Check your understanding

Exercise. A1

In [Example \(Checking estimability in a one-way layout\)](#), decide which of the following are estimable, using condition (c): $2\mu + \alpha_1 + \alpha_2$; $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4$; μ ; $4\mu + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$. For each estimable one give its least squares estimator in terms of the group means and its variance.

8.3.5.2 B. Practice

Exercise. B1

In [Example \(Checking estimability in a one-way layout\)](#), compute $\mathbf{H}_G$ for the drop-intercept generalized inverse, and identify the estimable function $\lambda^\top \mathbf{H}_G \beta$ that the software is really estimating when it reports “ $\hat{\alpha}_k$ ”. Do the same for the generalized inverse that deletes the row and column of α_4 .

Solution. With $\mathbf{G}$ holding $\text{diag}(1/n_1, \dots, 1/n_4)$ in its lower right block, $\mathbf{G}\mathbf{X}^\top\mathbf{X}$ has first row zero and row $k+1$ equal to $(1, \mathbf{e}_k^\top)$, because the row of $\mathbf{X}^\top\mathbf{X}$ for α_k is $n_k(1, \mathbf{e}_k^\top)$. So $E(\hat{\alpha}_k) = \mu + \alpha_k$: the reported “effect” estimates the group mean. If instead α_4 is deleted, the solution sets $\hat{\alpha}_4 = 0$, $\hat{\mu} = \bar{y}_4$ and $\hat{\alpha}_k = \bar{y}_k - \bar{y}_4$, so $E(\hat{\mu}) = \mu + \alpha_4$ and $E(\hat{\alpha}_k) = \alpha_k - \alpha_4$ for $k \leq 3$.

Exercise. B2

Let $\mathbf{\Lambda}^\top\boldsymbol{\beta}$ be estimable with $\mathbf{\Lambda}$ of full column rank q , and suppose $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ with $\sigma^2 > 0$. Show that $\text{Cov}(\mathbf{\Lambda}^\top\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{\Lambda}^\top\mathbf{G}\mathbf{\Lambda}$ is positive definite, whichever generalized inverse is used. *Hint:* write $\mathbf{\Lambda} = \mathbf{X}^\top\mathbf{P}$ and show $\mathbf{M}\mathbf{P}$ has full column rank.

8.3.5.3 C. Going deeper

Exercise. C1

Let $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ be nonestimable. Show that as $\mathbf{G}$ ranges over all generalized inverses of $\mathbf{X}^\top\mathbf{X}$, the number $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda}$ takes *every* real value, including negative ones. So the “variance” that software attaches to a nonestimable function can be anything at all.

Solution. Let $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ and $\mathbf{G}_0$ be one generalized inverse. By Proposition 1.3.3(e), $\mathbf{G} = \mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U}$ is a generalized inverse for every $\mathbf{U}$, and

$$\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + \mathbf{u}^\top\mathbf{U}\boldsymbol{\lambda}, \quad \mathbf{u} = (\mathbf{I} - \mathbf{G}_0\mathbf{A})^\top\boldsymbol{\lambda}.$$

Because $\boldsymbol{\lambda}$ is not estimable, $\boldsymbol{\lambda}^\top\mathbf{G}_0\mathbf{A} \neq \boldsymbol{\lambda}^\top$ (by (a) $\Leftrightarrow$ (e) of [Theorem 8.3.1](#)), so $\mathbf{u} \neq \mathbf{0}$. Also $\boldsymbol{\lambda} \neq \mathbf{0}$, since $\mathbf{0}$ is in $\mathcal{C}(\mathbf{X}^\top)$. Take $\mathbf{U} = t\mathbf{u}\boldsymbol{\lambda}^\top/(\|\mathbf{u}\|^2\|\boldsymbol{\lambda}\|^2)$. Then $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + t$, which takes every real value as t varies.

Exercise. C2

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that the minimum-norm least squares solution $\hat{\boldsymbol{\beta}}^+$ satisfies $E(\hat{\boldsymbol{\beta}}^+) = \mathbf{P}\boldsymbol{\beta}$, and that for any $\boldsymbol{\lambda}$, $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}}^+$ is the least squares estimator of the estimable function $(\mathbf{P}\boldsymbol{\lambda})^\top\boldsymbol{\beta}$. Interpret $\mathbf{P}\boldsymbol{\lambda}$ as the estimable function “closest” to $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$.